

Amir Evelio Hurtado Mena

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SUMMARY

Systems Engineer working on infrastructure and full-stack development for public institutions. I design and operate a production Kubernetes (OKD/OpenShift) cluster — virtual machines, networking, fault-tolerant database clusters and full observability — and build government-facing web platforms currently in active use.

EXPERIENCE

Full-Stack Developer & Infrastructure Engineer

Universidad Tecnológica de Pereira (UTP)

2025 – Present

Pereira, Colombia

- Develop web platforms and infrastructure for Colombian public institutions, in partnership with the Governments of Meta and Caldas.
- Lead infrastructure of **OKD — Meta**: a production OKD (OpenShift/Kubernetes) cluster for a network of public hospitals (ESEs) — see project below.
- Built **Expediente Municipal — Caldas**, in active use by the municipalities of Caldas; I continue to provide technical support.

PROJECTS

OKD — Meta · Government of Meta

In production

- Kubernetes (OKD/OpenShift) cluster built from scratch on virtual machines for a network of public hospitals (ESEs): each hospital runs local service containers, while the central datacenter runs 26 Spring Boot microservices and the Angular frontend.
- HA control plane (3 masters + 4 workers), bastion node with DNS (BIND9), DHCP and HAProxy load balancing; 3 isolated namespaces: production, pre-production and training.
- Fault-tolerant database clusters: 3-node MongoDB replica set and PostgreSQL cluster.
- Full observability: Prometheus with Kubernetes service autodiscovery, Grafana, ELK and Zipkin distributed tracing.
- Alternate datacenter (GTD, Medellín): a complete replication of the infrastructure with a single production namespace. Private network, VPN-only access.

OKD/OpenShift · Kubernetes · Linux · HAProxy · BIND9 · MongoDB · PostgreSQL · Spring Boot · Angular · Maven + JKube

Expediente Municipal — Caldas · Government of Caldas

In production

- Platform tracking the Territorial Planning Plan (POT) of the municipalities of Caldas, integrating documentation, indicators and cartography. In active use on a private institutional network (VPN).

Angular · Spring Boot · PostgreSQL · MongoDB

RISC-V Simulator · github.com/amirhurtado/RiscVSiriusStudio

In use at UTP

- Interactive RISC-V assembly simulator as a VS Code extension, built within the Sirius UTP research group: step-by-step visual execution of the datapath, registers and memory. Used to teach Computer Architecture at UTP.

TypeScript · React · VS Code API

PecesControl

In development

- Hardware and software platform for fish-farming monitoring: temperature and oxygen sensors and a food dispenser connected to an ESP32, with remote actions from the platform (dispense food, activate heaters, control water level).

ESP32 · React · Spring Boot

SKILLS

Infrastructure:	Kubernetes (OKD/OpenShift), Docker, Linux, HAProxy, BIND9 (DNS), DHCP, virtualization (VMs)
Observability:	Prometheus, Grafana, ELK (Elasticsearch, Logstash, Kibana), Zipkin
Databases:	MongoDB (replica sets), PostgreSQL (clusters)
Backend:	Spring Boot (Java), Node.js, Maven + JKube
Frontend:	TypeScript, React, Angular, Three.js, GSAP
Spoken:	Spanish (native), English (professional)

EDUCATION

B.Sc. Systems and Computer Engineering

Universidad Tecnológica de Pereira (UTP)

Graduated 2026

Pereira, Colombia